

SEMINAR 6 (Pilot). LIMITS OF FUNCTIONS. FIRST REMARKABLE LIMIT

1. Give formal definitions for the following notions. Construct the negation to each of them.

$$(a) \lim_{x \rightarrow a+0} f(x) = -\infty; \quad (b) \text{ (HW) } \lim_{x \rightarrow a-0} f(x) = L; \quad (c) \text{ (HW) } \lim_{x \rightarrow -\infty} f(x) = +\infty.$$

2. Find the following one-sided limits:

$$(a) \lim_{x \rightarrow +0} \frac{x}{|x|}; \quad (b) \lim_{x \rightarrow -0} \frac{x}{|x|}; \quad (c) \lim_{x \rightarrow 2+0} \frac{x^2 - x - 1}{x^2 - 5x + 6}; \quad (d) \lim_{x \rightarrow 2-0} \frac{x^2 - x - 1}{x^2 - 5x + 6}.$$

3. Suppose that $f(x) = \begin{cases} |1-x|, & x < 2 \\ \sqrt{x+2}, & x \geq 2 \end{cases}$ Find $\lim_{x \rightarrow 2-} f(x)$, $\lim_{x \rightarrow 2+} f(x)$. Does $\lim_{x \rightarrow 2} f(x)$ exist?

4. Find a and b such that $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$.

5. Find the limit $\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1}$ (m and n are positive integers).

6. (HW) Find $\lim_{x \rightarrow 1} \frac{\sqrt[m]{x} - 1}{\sqrt[n]{x} - 1}$ (m and n are positive integers). (*Hint:* Reduce this limit to the preceding one by an appropriate change of variables.)

7. (HW) Find the following one-sided limits:

$$(a) \lim_{x \rightarrow 7+0} \frac{|x-7|}{x^2 - 5x - 14}; \quad (b) \lim_{x \rightarrow 7-0} \frac{|x-7|}{x^2 - 5x - 14}; \quad (c) \lim_{x \rightarrow -1+0} \frac{(\sin x) + 1}{x + 1}; \quad (d) \lim_{x \rightarrow -1-0} \frac{(\sin x) + 1}{x + 1}.$$

8. (HW) Find a_i and b_i such that

$$(a) \lim_{x \rightarrow +\infty} \left(\sqrt{x^2 - x + 1} - a_1 x - b_1 \right) = 0; \quad (b) \lim_{x \rightarrow -\infty} \left(\sqrt{x^2 - x + 1} - a_2 x - b_2 \right) = 0.$$

THE FIRST REMARKABLE LIMIT:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

9. Find the following limits:

$$(a) \lim_{x \rightarrow 0} \frac{\sin 3x}{x}; \quad (b) \lim_{x \rightarrow 0} \frac{\sin 7x}{\sin 8x}; \quad (c) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}; \quad (d) \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a};$$

$$(e) \lim_{x \rightarrow 0} x \sin \frac{1}{x}; \quad (f) \lim_{x \rightarrow \pi} \frac{\sin x}{\pi^2 - x^2}; \quad (g) \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x}{1 + \sin px - \cos px}; \quad (h) \lim_{x \rightarrow +\infty} \left(\sin \sqrt{x+1} - \sin \sqrt{x} \right).$$

10. (HW) Evaluate the following limits:

$$(a) \lim_{x \rightarrow 0} \frac{\operatorname{tg} 5x}{\sin 2x}; \quad (b) \lim_{x \rightarrow \frac{\pi}{2}} \left(\cos x \cdot \sin \frac{2}{2x - \pi} \right); \quad (c) \lim_{x \rightarrow 0} \frac{1 - \cos 8x}{5x^2};$$

$$(d) \lim_{x \rightarrow 0} \frac{\sqrt{3} - \sqrt{x+3}}{\sin 2x}; \quad (e) \lim_{x \rightarrow 0} \frac{\cos \alpha x - \cos \beta x}{x \operatorname{tg} \gamma x}; \quad (f^*) \lim_{x \rightarrow 0} \frac{x^2}{\sqrt{1+x \sin x} - \sqrt{\cos x}}.$$